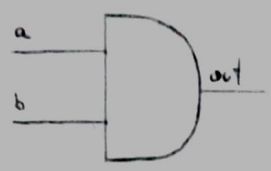
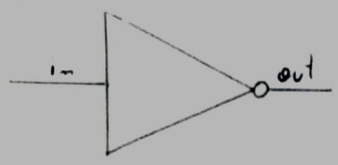
- AND



a	b	out
0	0	0
0	1	0
1	0	0
1	1	1

Not (Nand (a,b))

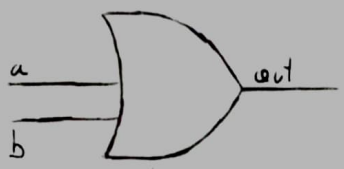
- NOT



in	out
1	0
0	1

Nand (a,a)

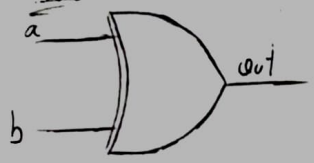
- OR



a	b	out
0	0	0
0	1	1
1	0	1
1	1	1

Nand (Not(a) , Not(b))

- XOR

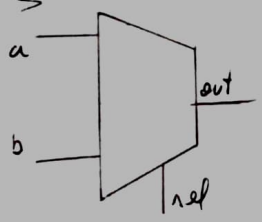


a	b	out
0	0	0
0	1	1
1	0	1
1	1	0

~~Not(a) AND Not(b) OR Not(b) AND a~~

Not(a) AND b OR a AND Not(b)

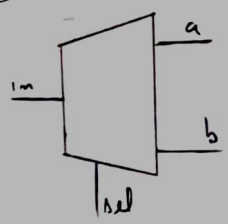
- Mux



sel	out
0	a
1	b

a AND Not(sel) OR b AND sel

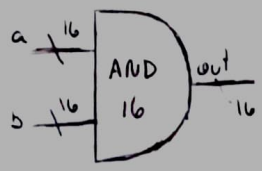
- DMux



sel	out
0	a
1	b

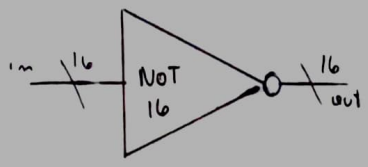
in AND Not(sel) -> out a

in AND sel -> out b



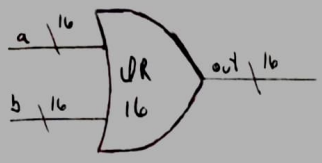
a[i] AND b[i] -> out[i]

i indo de 0 a 15



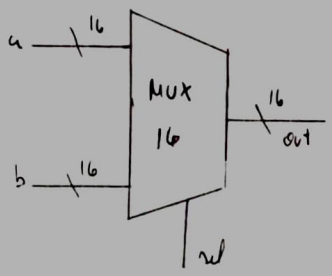
NOT in[i] -> out[i]

i indo de 0 a 15



a[i] OR b[i] -> out[i]

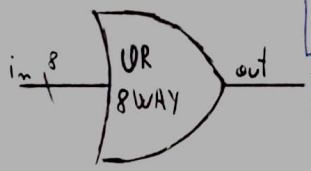
i indo de 0 a 15



sel = 0 -> a[i] = out[i]

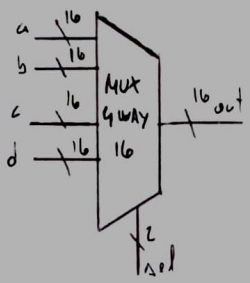
sel = 1 -> b[i] = out[i]

i indo de 0 a 15



[(in[0] OR in[1]) OR (in[2] OR in[3])] OR

[(in[4] OR in[5]) OR (in[6] OR in[7])]

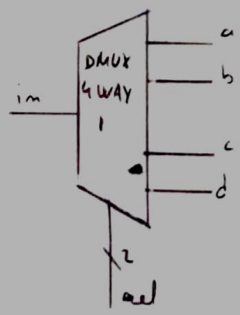


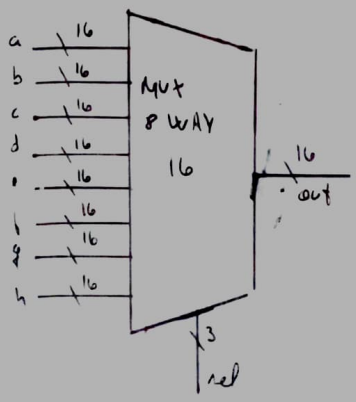
sel = (0,0) -> out = a

sel = (0,1) -> out = b

sel = (1,0) -> out = c

sel = (1,1) -> out = d





sel (0,0,0) -> out = a

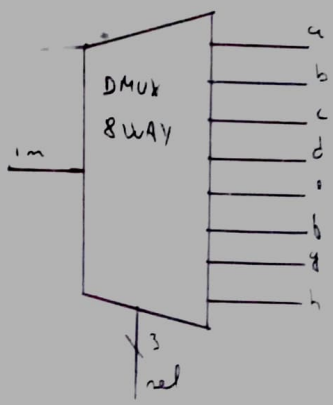
sel (0,0,1) -> out = b

sel (0,1,0) -> out = c

sel (0,1,1) -> out = d

sel (1,0,0) -> out = e

sel (1,0,1) -> out = f

sel (1,1,0) -> out = g

sel (1,1,1) -> out = h